

We show an algorithm to check the Tortkara identity

Tortkara identity is given by

$$(x*y)*(z*y)=((x*y)*z + (y*z)*x + (z*x)*y) * y$$

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> restart:with(linalg):
> with(ListTools):
> with(combinat):
>
> law:=proc(i,j)
> if (i,j)=(1,1) then return c111*e[1]+c112*e[2]; end if;
> if (i,j)=(1,2) then return c121*e[1]+c122*e[2]; end if;
> if (i,j)=(2,1) then return c211*e[1]+c212*e[2]; end if;
> if (i,j)=(2,2) then return c221*e[1]+c222*e[2]; end if;
> end proc:
>
> prod:=proc(u,v,n)
> local exp;
> exp:=0;
> for i from 1 to n do
> for j from 1 to n do
> exp:=exp + coeff(u,e[i])*coeff(v,e[j])*law(i,j);
> end do; end do; return exp;
> end proc:
Warning, (in prod) `i` is implicitly declared local
Warning, (in prod) `j` is implicitly declared local
> tortkara_parameters:=proc(n)
> local L,M,N,P;L:=[];M:=[];N:=[];P:=[];
> for i from 1 to n do
> L:=[op(L),i,i,i]; end do;
> M:=permute(L,3);
> for j from 1 to nops(M) do
> eq[j]:=prod(prod(e[M[j][1]],e[M[j][2]],n),prod(e[M[j][3]],e[M[j]
[2]],n),n)-prod(prod(prod(e[M[j][1]],e[M[j][2]],n),e[M[j][3]],n)+
prod(prod(e[M[j][2]],e[M[j][3]],n),e[M[j][1]],n)+prod(prod(e[M[j]
[3]],e[M[j][1]],n),e[M[j][2]],n),e[M[j][2]],n);
> end do;
> N:=[seq(eq[k], k=1..nops(M))];
> for k from 1 to nops(N) do for h from 1 to n do
> P:=[op(P),coeff(N[k],e[h])];
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> end do; end do; P:=[op(P), c111^2+c112^2<>0, c121^2+c122^2<>0,
  c211^2+c212^2<>0, c221^2+c222^2<>0];
> return M, factor(N), solve(P);
> end proc;
```

Warning, (in tortkara_parameters) `i` is implicitly declared local
 Warning, (in tortkara_parameters) `j` is implicitly declared local
 Warning, (in tortkara_parameters) `eq` is implicitly declared local
 Warning, (in tortkara_parameters) `k` is implicitly declared local
 Warning, (in tortkara_parameters) `h` is implicitly declared local

```
> tortkara_parameters(2);
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$$\begin{aligned}
 & [[1, 1, 1], [1, 1, 2], [1, 2, 1], [1, 2, 2], [2, 1, 1], [2, 1, 2], [2, 2, 1], [2, 2, 2]], \left[-2 c11l^3 e_1 \right. \\
 & - 2 c11l^2 c112 e_2 + c11l c112 c12l e_1 + c11l c112 c122 e_2 - 5 c11l c112 c211 e_1 \\
 & - 2 c11l c112 c212 e_2 - 3 c112^2 c211 e_2 + c112^2 c221 e_1 + c112^2 c222 e_2 \\
 & - 3 c112 c211 c212 e_1 - 3 c112 c212^2 e_2, -2 c11l^2 c12l e_1 - 2 c11l c112 c12l e_2 \\
 & - c11l c112 c221 e_1 + c11l c12l c212 e_1 - 2 c11l c122 c211 e_1 - c11l c211 c212 e_1 \\
 & - c112^2 c221 e_2 - c112 c12l c211 e_1 - c112 c12l c212 e_2 - c112 c122 c211 e_2 \\
 & - c112 c211 c212 e_2 - c112 c211 c222 e_1 + c112 c212 c221 e_1 - c122 c211 c212 e_1 \\
 & - c122 c212^2 e_2 - c211 c212^2 e_1 - c212^3 e_2, -c11l c12l^2 e_1 - 2 c11l c12l c122 e_2 \\
 & - c11l c12l c211 e_1 - c11l c122 c211 e_2 - c11l c122 c221 e_1 - c11l c122 c222 e_2 \\
 & + c112 c12l^2 e_2 - 2 c112 c12l c221 e_1 - c112 c12l c222 e_2 - c112 c122 c221 e_2 \\
 & - c112 c211 c221 e_1 - c112 c211 c222 e_2 - c112 c221 c222 e_1 - c112 c222^2 e_2 \\
 & + c12l^2 c122 e_1 + c12l c122^2 e_2 + c12l c122 c212 e_2 - c12l c211 c212 e_1 - c122^2 c211 e_2 \\
 & + c122^2 c221 e_1 + c122^2 c222 e_2 - c122 c211 c212 e_2 - c122 c212 c221 e_1 \\
 & - c122 c212 c222 e_2 - c212^2 c221 e_1 - c212^2 c222 e_2, -c11l c122 c221 e_2 \\
 & + c112 c12l c221 e_2 - c112 c22l^2 e_1 - c112 c221 c222 e_2 - c12l^3 e_1 - c12l^2 c122 e_2 \\
 & - c12l^2 c211 e_1 + c12l^2 c222 e_1 - c12l c122 c211 e_2 - 2 c12l c122 c221 e_1 \\
 & - c12l c211 c222 e_1 - c12l c212 c221 e_1 - c122^2 c221 e_2 - 2 c122 c211 c222 e_2 \\
 & - 2 c212 c221 c222 e_1 - 2 c212 c222^2 e_2, -2 c11l^2 c12l e_1 - 2 c11l c112 c12l e_2 \\
 & - c11l c112 c221 e_1 - 2 c11l c122 c211 e_1 - c11l c122 c212 e_2 + c11l c212^2 e_2 \\
 & - c112^2 c221 e_2 - c112 c12l c212 e_2 - c112 c21l^2 e_1 - 2 c112 c211 c212 e_2 \\
 & \left. - c112 c211 c222 e_1 + c112 c212 c221 e_1 - c122 c211 c212 e_1 - c122 c212^2 e_2 \right] \quad (1)
 \end{aligned}$$

$$\begin{aligned}
& -c211c212^2e_1 - c212^3e_2 - c111l^2c221e_1 - c111c112c221e_2 - c111c12l^2e_1 \\
& - c111c12l c211e_1 - c111c122c221e_1 + c111c21l^2e_1 - c111c211c222e_1 \\
& - c111c212c221e_1 - c112c12l^2e_2 - c112c12l c211e_2 - c112c122c221e_2 \\
& + c112c21l^2e_2 - c112c211c221e_1 - c112c211c222e_2 - 2c112c212c221e_2 \\
& - c12l c122c211e_1 - c12l c122c212e_2 + c12l c211c212e_1 - c122c21l^2e_1 \\
& - c122c211c222e_1 - c122c212c222e_2 + c21l^2c212e_1 + c211c212^2e_2 \\
& - 2c211c212c222e_1 + c212^2c221e_1 - c212^2c222e_2 - c111c122c221e_2 \\
& + c112c12l c221e_2 - c112c22l^2e_1 - c112c221c222e_2 - c12l^3e_1 - c12l^2c122e_2 \\
& - c12l^2c211e_1 - c12l c122c211e_2 - c12l c122c221e_1 - c12l c122c222e_2 \\
& - c12l c212c221e_1 + c12l c212c222e_2 - c122c211c221e_1 - 2c122c211c222e_2 \\
& - c122c212c221e_2 - 2c212c221c222e_1 - 2c212c222^2e_2, c111c22l^2e_1 + c112c22l^2e_2 \\
& - 3c12l^2c221e_1 - 3c12l c122c221e_2 - 2c12l c221c222e_1 - 3c122c22l^2e_1 \\
& - 5c122c221c222e_2 + c211c221c222e_1 + c212c221c222e_2 - 2c221c222^2e_1 \\
& - 2c222^3e_2], \left\{ c111 = -c212, c112 = -\frac{c222c212}{c221}, c12l = -c222, c122 = -\frac{c222^2}{c221}, c211 \right. \\
& \left. = \frac{c212c221}{c222}, c212 = c212, c221 = c221, c222 = c222 \right\}
\end{aligned}$$

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